

Project 02: NEM Network Constraints & Price Divergence Intelligence System

Combined visual report for NotebookLM. This report brings together all Plotly chart outputs from the project into one visual reference document.

STUDY PERIOD

Feb 2026

2026-02-01 to 2026-03-01

FOCUS REGIONS

NSW1 / VIC1

Regional price comparison

NSW1 SPIKES

53

RRP above \$300/MWh

DIVERGENCE EVENTS

307

Spread above \$100/MWh

Visual Storyline

The charts are designed to explain how market price behaviour, network constraints, interconnector flow, and regional price divergence connect to spike classifications and decision intelligence.

- NSW1 recorded spike intervals while VIC1 did not breach the selected \$300/MWh threshold.
- Constraint activity was elevated during NSW1 spike intervals, but marginal value and violation signals did not explain every spike on their own.
- Broader interconnector congestion-style signals were present during most NSW1 spike intervals.
- NSW1-VIC1 price divergence was material in concentrated periods, supporting the network-driven market interpretation.

1. RRP By Region With Spike Markers

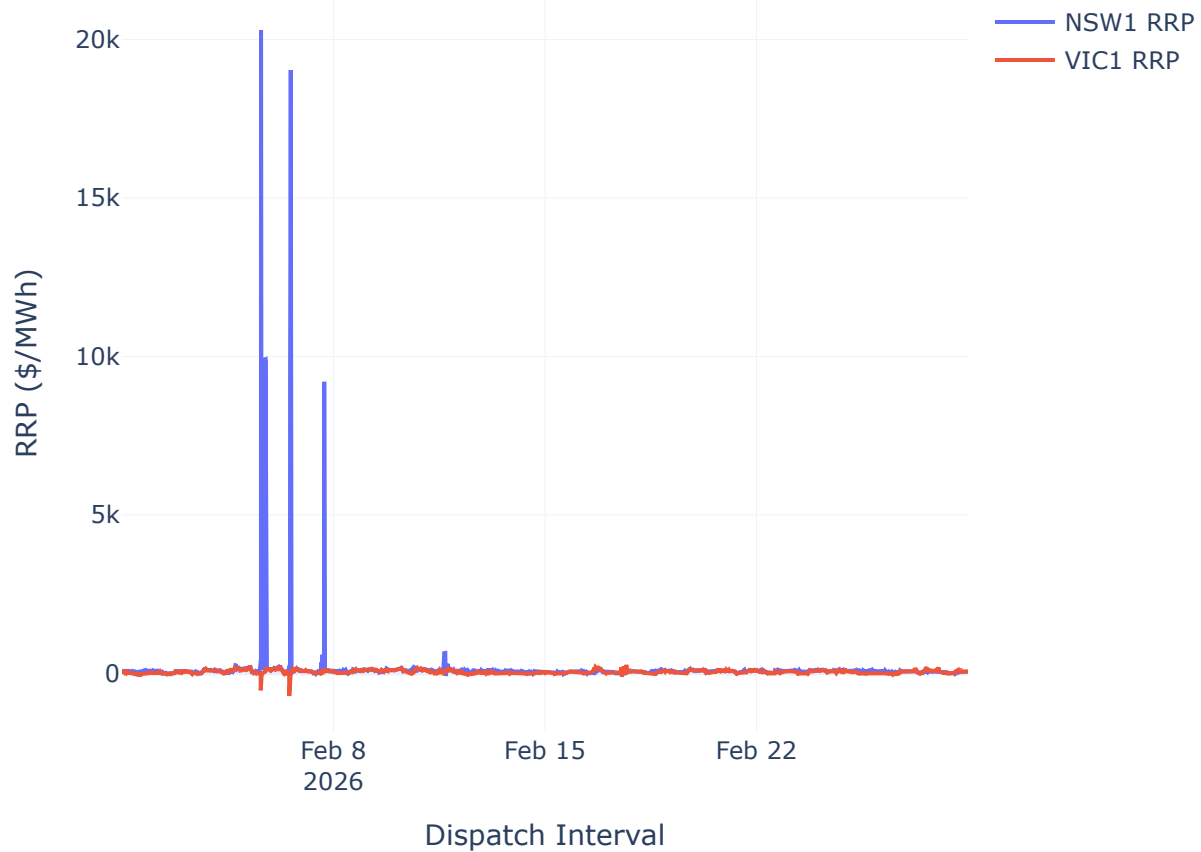
Shows NSW1 and VIC1 RRP over time with price spike intervals highlighted.

WebGL is not supported
by your browser - visit
<https://get.webgl.org> for
more info

2. NSW1 Vs VIC1 RRP Comparison

Compares regional prices on the same time axis to show when prices move together or separate.

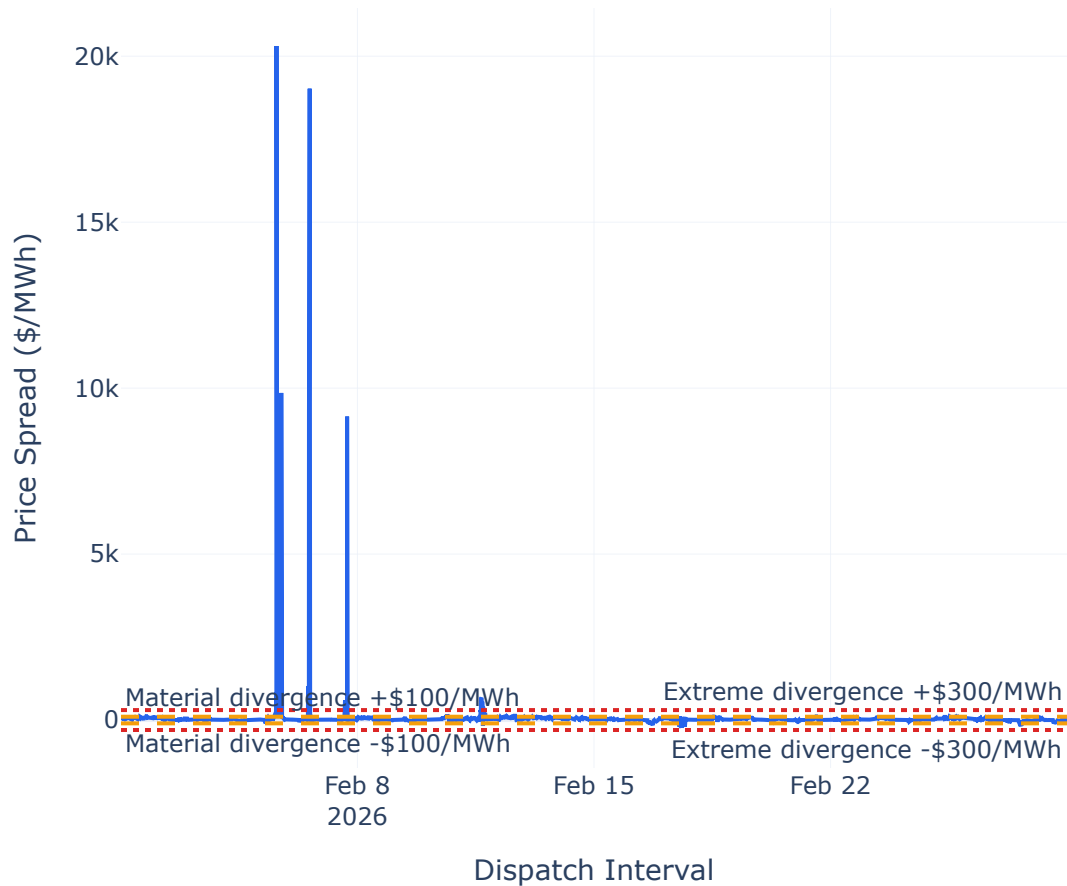
NSW1 vs VIC1 Regional Reference Price



3. NSW1-VIC1 Price Spread

Displays signed price spread and divergence thresholds for material and extreme separation.

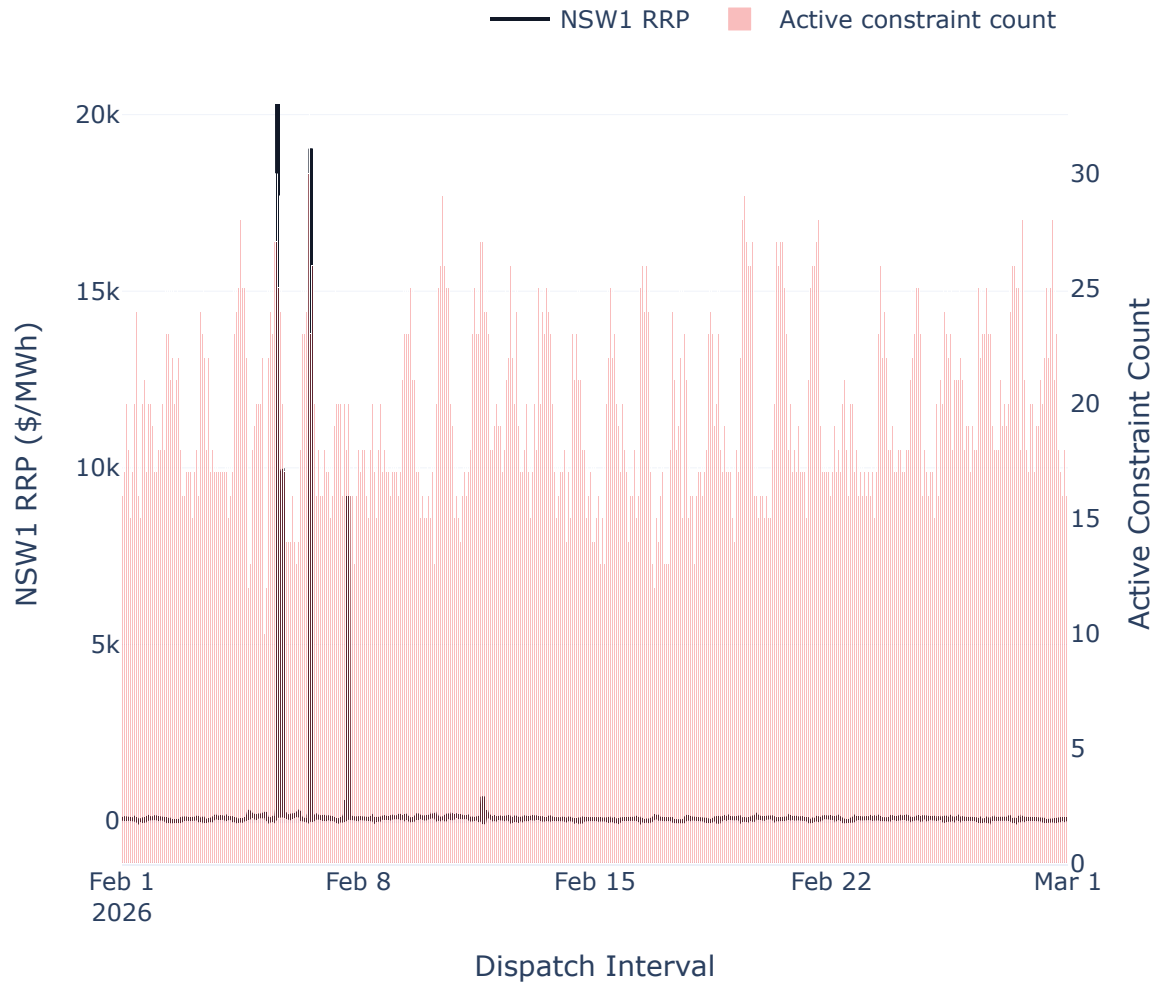
NSW1-VIC1 Price Spread Over Time



4. Constraint Activity Vs NSW1 RRP

Compares NSW1 RRP against active constraint count to test whether spike periods align with higher constraint activity.

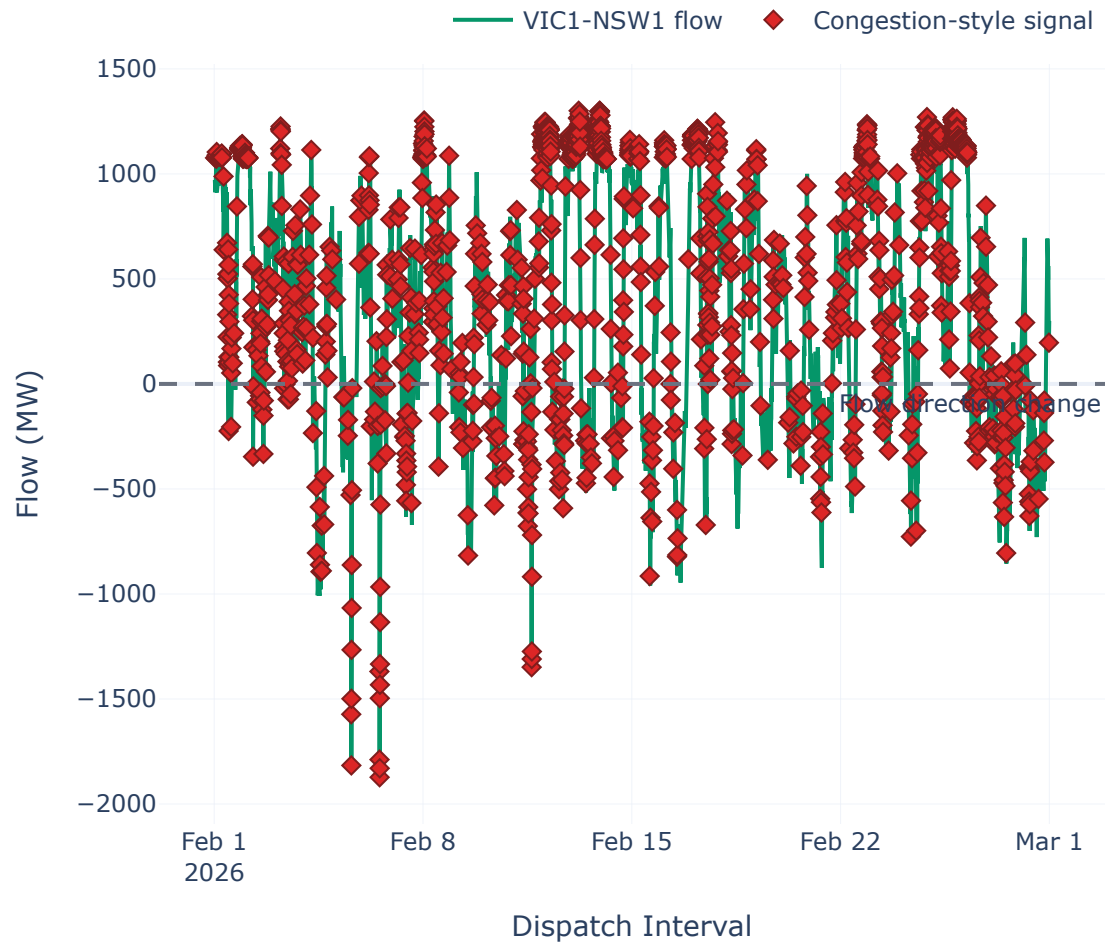
NSW1 RRP And Active Constraint Count



5. VIC1-NSW1 Interconnector Flow

Shows direct NSW-VIC interconnector flow behaviour and congestion-style signal intervals.

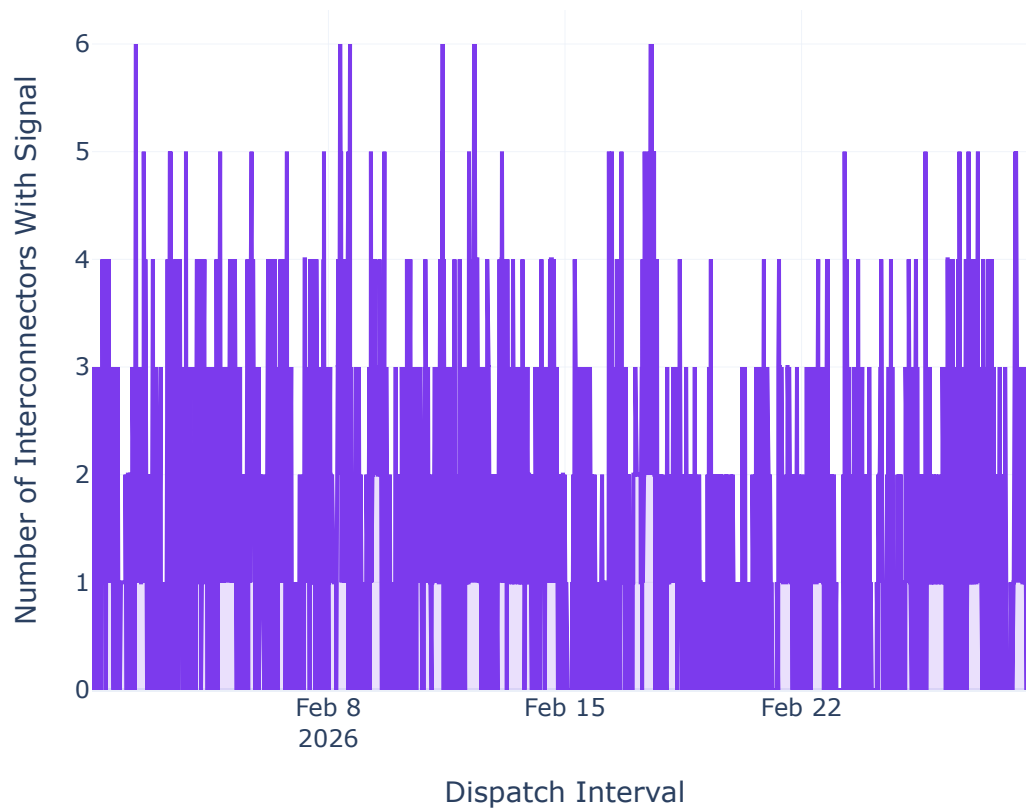
VIC1-NSW1 Interconnector Flow And Congestion-Style Signals



6. Congestion Signal Count

Shows broader interconnector stress by counting how many interconnectors had congestion-style signals at each interval.

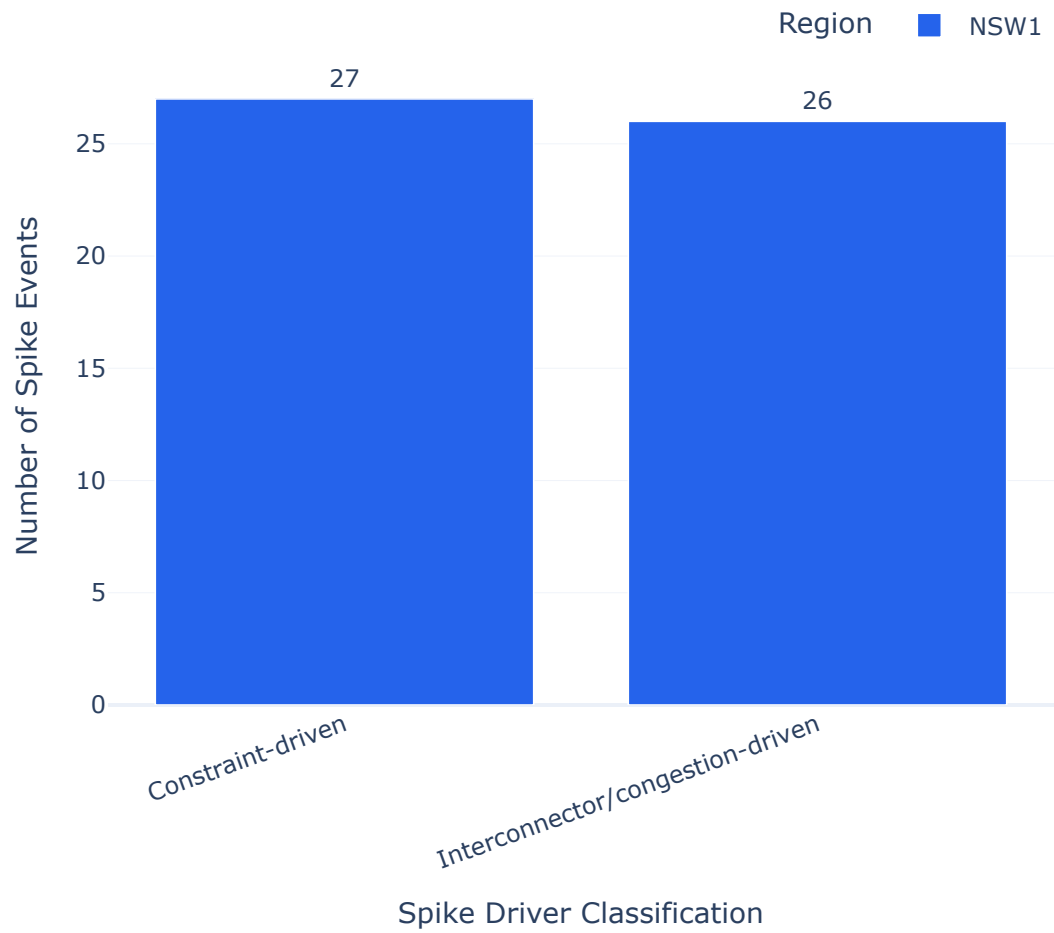
Interconnector Congestion-Style Signal Count



7. Spike Classification Summary

Summarises classified spike events by likely market driver.

Price Spike Classification Summary

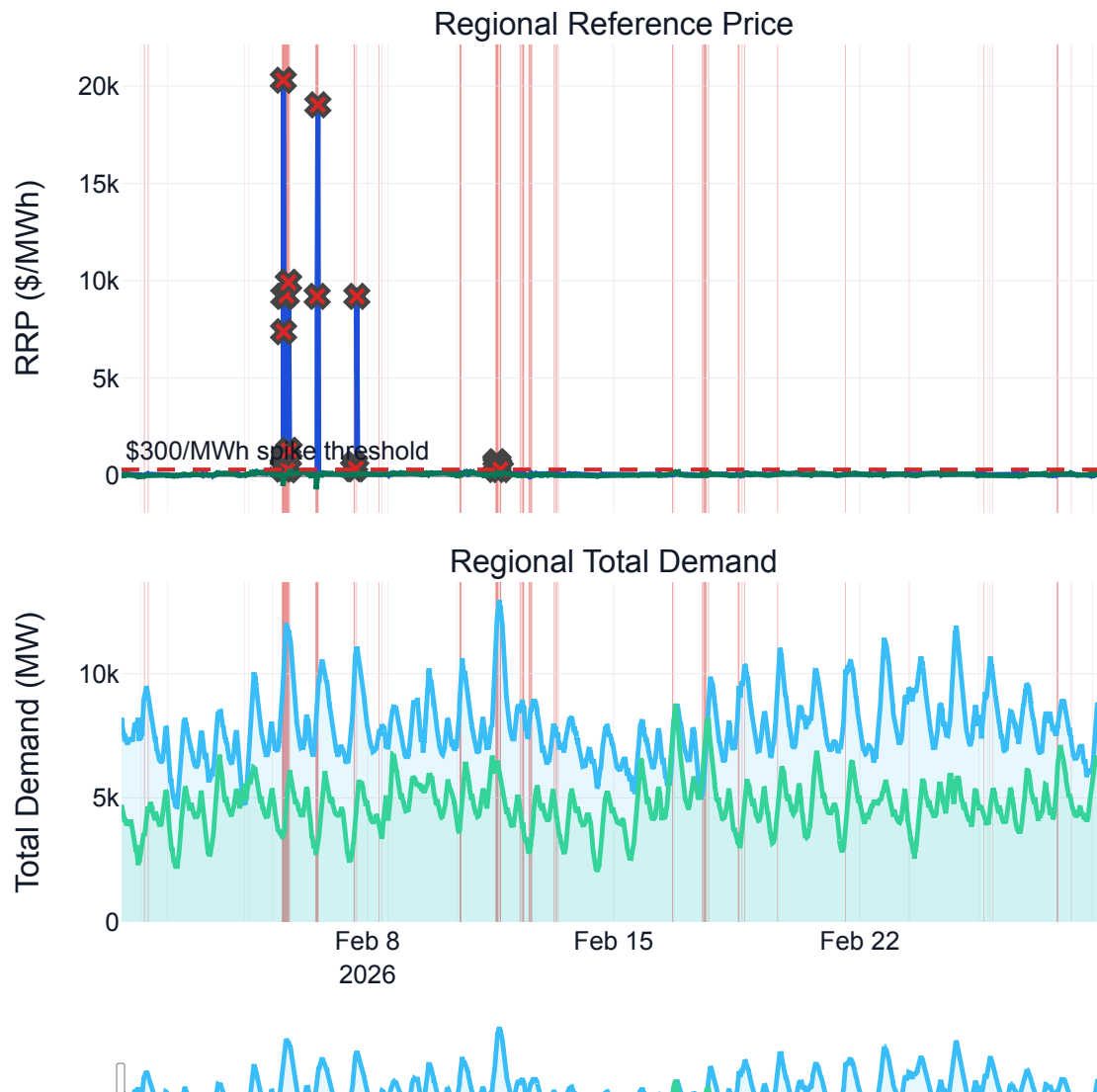


8. Demand And Price Overview

Compares RRP and demand between NSW1 and VIC1, including spike markers and divergence periods where available.

Interactive Demand And Price Overview: NSW1 vs VIC1

RRP, total demand, spike markers, and divergence periods
NSW1 RRP, NSW1 spike, NSW1 Demand
VIC1 RRP, VIC1 Demand



Final Analyst Message

Project 02 demonstrates an end-to-end network intelligence workflow for the NEM. It starts with cleaned dispatch data, adds market and network features, classifies price spikes by

likely driver, and translates those classifications into decision recommendations. The visual pack supports slide creation, study review, and dashboard design.

Generated from local Project 02 Plotly chart outputs. Interactive HTML charts are stored under [outputs/charts/](#).